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REVIEW ARTICLE

Cobalt Electroplating Interconnects in Advanced Chip Fabrication: A Review

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Abstract

Electroplating is a core technology in high-end chip manufacturing, with its technical challenges centered on achieving void-free filling of metal coatings within micro- and nanoscale trenches and vias. Copper, the current mainstream interconnect material, can no longer meet the performance and reliability demands posed by the continuous reduction in interconnect dimensions. In contrast, cobalt offers superior electrical properties and safety at the micro-/nanoscale, making it a promising next-generation interconnect material to replace copper. This paper provides a comprehensive overview of cobalt interconnect technology in chip manufacturing. It examines the electrochemical behavior and interactions of cobalt ions, boric acid, and additives, summarizing the cobalt filling effects and associated mechanism models under different process compositions. The discussion also highlights advanced in situ electrochemical characterization technologies to investigate the synergistic and antagonistic effects of various plating bath components, the influence of pH, temperature, and additives on cobalt nucleation and growth, and the surface, interface, and filling processes during cobalt electrodeposition in micro-/nanostructures. Furthermore, the paper explores annealing treatments for cobalt coatings, emphasizing their role in optimizing performance.

Keywords Chip fabrication · cobalt interconnects · additives · mechanism · surface and interface process · annealing treatment

Introduction

The rapid advancement of electronic information and technology has significantly enhanced living standards, driven scientific and technological innovation, and promoted social progress. At the heart of electronic information and technology lies the design, manufacturing, and testing of chips. Electroplating plays a pivotal role in the fabrication, integration, and packaging of integrated circuit chips. It is the only technology capable of achieving nanoscale electronic logic interconnects and enabling the large-scale, controllable manufacturing of metal-based micro-/nanostructures.[1–6] Furthermore, electroplating serves as a foundation for semiconductor chip interconnects and holds promise for the future integration and packaging of carbon-based, biological, and quantum chips.

Copper has high resistance to electromigration and excellent thermal and electrical conductivity, so it is the main metal material for electrical interconnects,[7–17] as shown in Fig. 1. However, with the rapid development of integrated circuit industry, the wiring density and metal layer stacking continually increase, e.g., the length of the chip internal metal wiring has been up to tens of kilometers, and the number of the metal wiring layers is up to 15. The improvement in energy consumption and performance requires the size of transistors and metal interconnects to shrink continuously, with the resulting size effects. The more significant grain boundary scattering and interface scattering of electrons inside the metal caused by size effects are shown in Fig. 2.[13] When the width of copper interconnects line decreases, the current load increases sharply, then the resistivity increases

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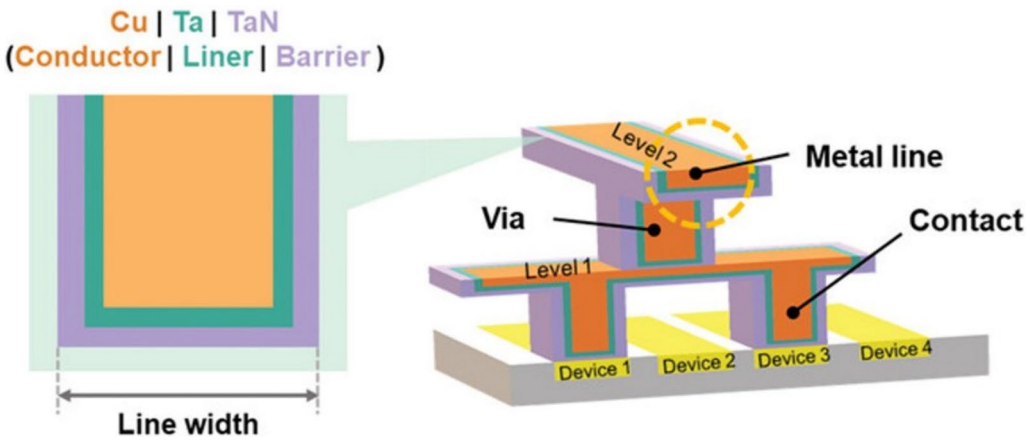


Fig. 1. Typical metallization of a copper interconnects. Reprinted with permission under a Creative Commons CC BY License from Ref. 13, Copyright 2023, Wiley-VCH.

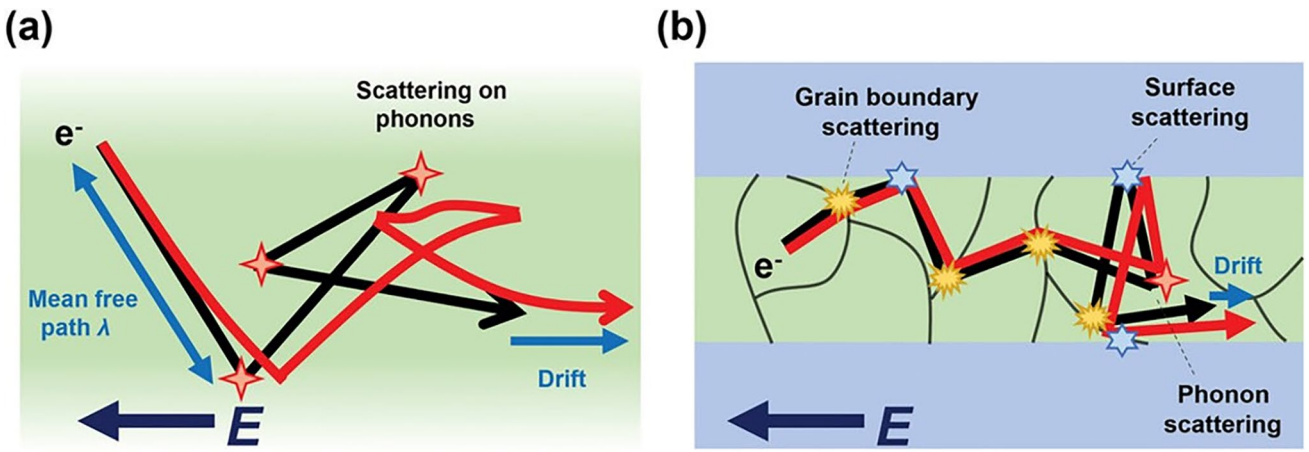


Fig. 2. Scattering mechanism on (a) bulk and (b) line or thin film at the nanoscale. Reprinted with permission under a Creative Commons CC BY License from Ref. 13, Copyright 2023, Wiley-VCH.

sharply, which will damage the chip performance due to the serious resistance and capacitance delay and a large amount of Joule heat. Therefore, copper interconnects have been unable to meet the development needs of advanced integrated circuit in the post-Moore era, and it is urgent to find next-generation metal interconnect materials.

The electron mean free path of copper is 39 nm.[18] Generally, the larger the electron mean free path of the bulk metal, the lower the resistivity of the metal. When the characteristic size of interconnects is reduced to the electron mean free path of copper, the electrons inside the copper are subjected to strong grain boundary scattering and interface scattering, causing a sudden increase in resistivity. Compared with copper, cobalt has a smaller electron mean free path (about 10 nm);[18] that is, cobalt possesses a smaller size effect than copper.[19] When the trench critical dimension is above about 7 nm, copper wires have lower resistivity. When the trench critical dimension is below about 7 nm, cobalt wires exhibit even lower resistivity.[20] The resistivities of different widths for Cu and Co are shown in Fig. 3. In addition, in copper interconnects, the TiN barrier layer and copper seed layer with about 2 nm thickness in total are needed to prevent the interdiffusion of copper and dielectric Si/SiO_2, which further reduces the width of copper interconnect line, resulting in a higher resistivity. Cobalt itself can act as a barrier layer or seed layer.[21] Theoretically, using cobalt as interconnect metal does not require a barrier layer and a seed layer with higher resistivity, thereby further improving the conductivity of the interconnect metal. Although cobalt has better electromigration resistance than copper, in practice, if a barrier layer is not used, cobalt will still diffuse into the SiO_2 dielectric layer or porous low-K dielectric layer.[22] In the absence of an seed layer or barrier layer, cobalt will also diffuse into the trench of other layers along the interface of the filling

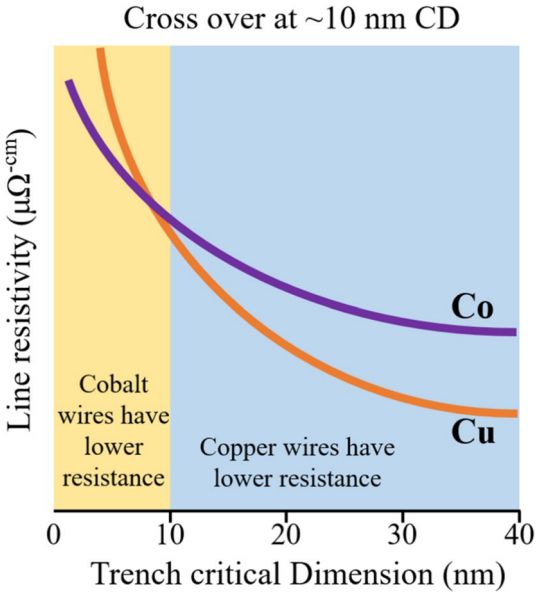


Fig. 3. Resistivity as a function of trench critical dimension$_{metal}$ for Cu and Co. Data from Ref. 20.

layer/dielectric layer.[23] Cobalt is an ideal next-generation interconnected metal material to replace copper.

Cobalt interconnects still use classical damascene process. As shown in Fig. 4, after the via or trench is etched, the barrier layer TiN and the adhesion layer WC are fabricated by chemical vapor deposition (CVD) and atomic layer deposition, respectively. The cobalt seed layer has been formed by CVD, and, subsequently, the electroplating technology has been used to realize the cobalt superfilling. Finally, chemical mechanical polishing (CMP) makes the rough surface flat. Cobalt electroplating is one of the key technologies to achieve high quality interconnects.

Nowadays, the Taiwan Semiconductor Manufacturing Company and Intel Corporation have applied cobalt electroplating interconnects to advanced processes in the 7- nm and 10-nm nodes, respectively.[24–28] Figure 5 shows the corresponding cobalt electroplating filling process. However, the process of advanced cobalt electroplating interconnects is relatively difficult due to the negative electro-reduction potential of cobalt ions and the more violent hydrogen evolution reaction; therefore, there are only a few reports about cobalt electroplating filling. The reaction process of cobalt electroplating and the mechanism of each component need to be further explored.

In this paper, the current research progress of cobalt electroplating, the role of each component, and the mechanism of superfilling are reviewed, and the basic research and technology development direction of cobalt electroplating are prospected.

Cobalt Electroplating Bath

A cobalt electroplating bath is mainly composed of virgin make-up solution (VMS) and organic additives. The VMS consists of cobalt sulfate heptahydrate ($CoSO_4{\cdot}7H_2O$), boric acid (H_3BO_3), cobalt chloride hexahydrate ($CoCl_2{\cdot}6H_2O$), and sulfuric acid (H_2SO_4). Table I presents the composition of the VMS.[29–31] $CoSO_4{\cdot}7H_2O$ is the metal source of cobalt electrodeposition. If the content of cobalt ions is too high,

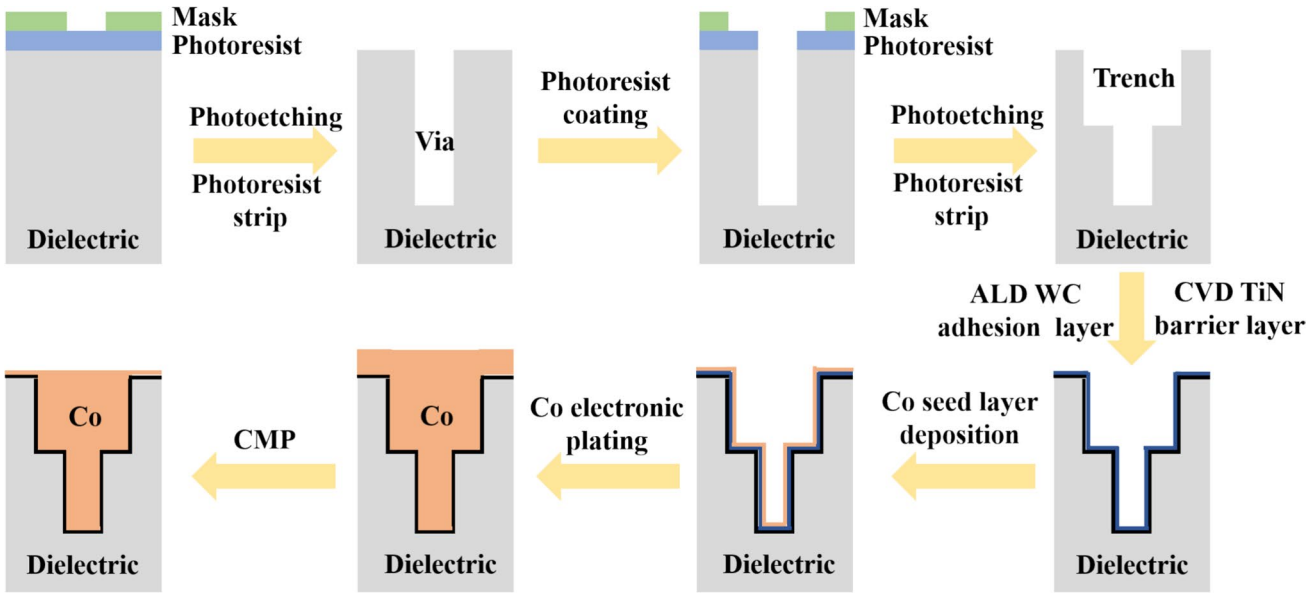


Fig. 4. Cobalt interconnect damascene process.

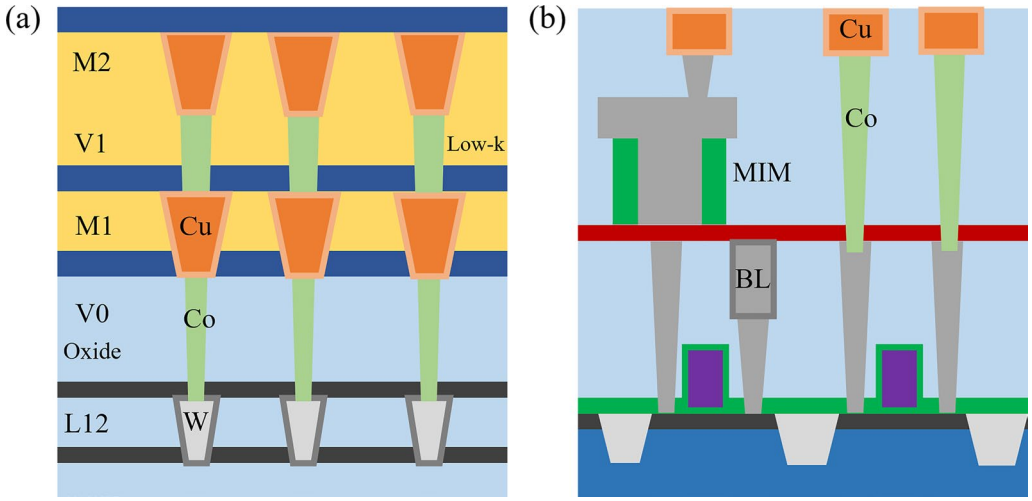


Fig. 5. (a) Cross-sectional schematic of Co bottom-up contact fill and via prefill, and (b) gate contact integrated with local Co interconnect at the back end of Intel's 10-nm process.

Table 1. Composition of the VMS for cobalt electroplating.

Chemicals	Contents	Functions	References
$CoSO_4 \cdot 7H_2O$	0.03–0.5 M	Main salt	[29–31]
$CoCl_2 \cdot 6H_2O$	0–0.84 mM	Main salt and activator	[29–31]
H_3BO_3	0.25–0.65 M	Buffering agent	[29–31]
H_2SO_4	A few drops	pH regulator (3–4)	[29–31]

the surface roughness of coating will increase. In contrast, if the content is too small, it is easy to produce intermediate inclusions and surface defects of the coating.[29] $CoCl_2 \cdot 6H_2O$ can provide chloride ions. On the one hand, they can promote the dissolution of cobalt anode during electroplating, while, on the other hand, they can cooperate with organic additives to change the cobalt reaction process.[30–32] H_2SO_4 regulates the pH value of the electroplating solution to 3–4 and prevents the electrodeposited cobalt layer from passivation and formation of $Co(OH)_2$, which makes the reaction difficult to continue. Since the reduction potential of cobalt is more negative than that of hydrogen, the hydrogen evolution reaction often occurs simultaneously with the cobalt ions electro-reduction during cobalt electroplating. To avoid a large change in the local pH value, it is often necessary to add a pH buffering agent to maintain a constant pH value. H_3BO_3, as a pH buffering agent, can prevent the pH changes caused by hydrogen evolution reactions. Organic additives are the key components to achieve cobalt bottom-up superfilling.[33–38]

It is worth noting that the mechanism H_3BO_3 in the electroplating bath is not yet clear, despite it being able to achieve a pH buffering effect due to the dynamic balance of protonation and deprotonation by forming polyborates. Rigsby et al.[39] found that H_3BO_3 could refine grain and reduce coating roughness, as shown in Fig. 6. Furthermore, when the content of H_3BO_3 was too low, the trenches were sealed in advance, leaving many holes and gaps, as displayed in Fig. 7. The H_3BO_3 plays a relatively complex role in cobalt electroplating interconnects. During the cobalt electroplating process, due to the continuous occurrence of the hydrogen evolution reaction, the pH value of the local electrolyte near the cathode surface increases significantly. The H_3BO_3 must be converted into polyborates and the released H^+ ions will reduce the local pH value and hinder the formation of $Co(OH)_2$ near the cathode surface.[40] In addition, polyborates are easily adsorbed on the cathode surface, occupying active sites for atomic hydrogen electro-reduction and inhibiting the hydrogen evolution reaction.[41] Generally, most chemical substances are adsorbed on the electrode surface, which can inhibit the electrodeposition of metal ions. However, as the concentration of H_3BO_3 increases, the adsorbed polyborate results in an increase in the current utilization rate of cobalt electrodeposition.[39] That is to say, the mechanism of converting H_3BO_3 into polyborates on the cathode surface, the mechanism of inhibiting the reduction of atomic hydrogen, the types of polyborates adsorbed on the cathode surface and their adsorption modes, and the mechanism of polyborates capturing metal ions to form complexes to improve current utilization are currently inconclusive. Research on H_3BO_3 in cobalt electroplating interconnects is still in its preliminary stage and there is room for discussion on the mechanism, but what is certain is that H_3BO_3 plays an indispensable role in cobalt electroplating from three aspects: stabilizing the pH, inhibiting surface hydrogen evolution, and promoting cobalt electrodeposition for high current utilization.

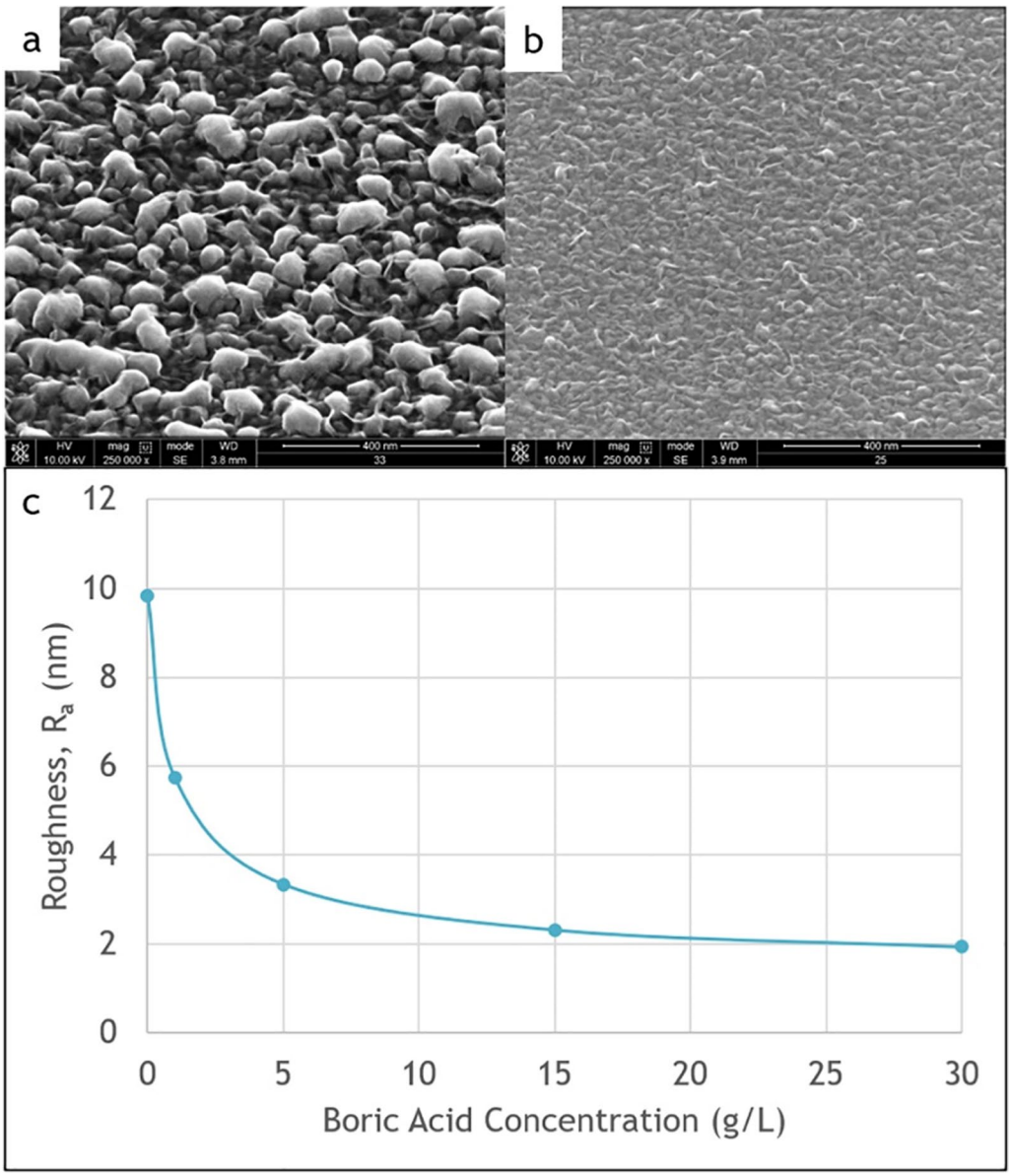


Fig. 6. Top-down SEM images of cobalt films with (a) 0 g/L, (b) 30 g/L H_3BO_3, and (c) AFM roughness as a function of H_3BO_3 concentration. Reprinted with permission from Ref. 39, Copyright 2020, IOP Publishing

Additives and Superfilling Mechanism

Metal cobalt is an iron group element, so the additives suitable for electroplating of iron group metals can be used for cobalt electroplating. Alkynol/enol compounds (including sulfonated compounds) are typical additives for cobalt electrodeposition. The molecular structure is shown in Fig. 8. However, to date, there are almost no reports on the use of alkynol/enol compounds for bottom-up superfilling of cobalt, and only a few studies have explored their reaction process on the surface of cobalt electrodes. Jin et al.[42] showed the molecule structure, adsorption ability, and electrochemical property of butynediol. Pan et al.[43] believed that the electrodeposited cobalt can catalyze the occurrence of the hydrogen evolution reaction, and the atomic hydrogen produced continuously reduced butynediol adsorbed on the cobalt electrode surface. Kiruba et al.[36] reported that butynediol can be reduced to butenediol on the cobalt electrode surface in the electroplating bath through Raman spectroscopy. In addition, there was electrochemical oscillation behavior in the initial stage of cobalt electrodeposition due to the three reasons of the interaction between atomic hydrogen and butynediol adsorption, the reaction between atomic hydrogen and butynediol, and the cobalt ions electro-reduction. Although there are no reports on the use of alkynol/enol compounds for cobalt bottom-up superfilling, this does

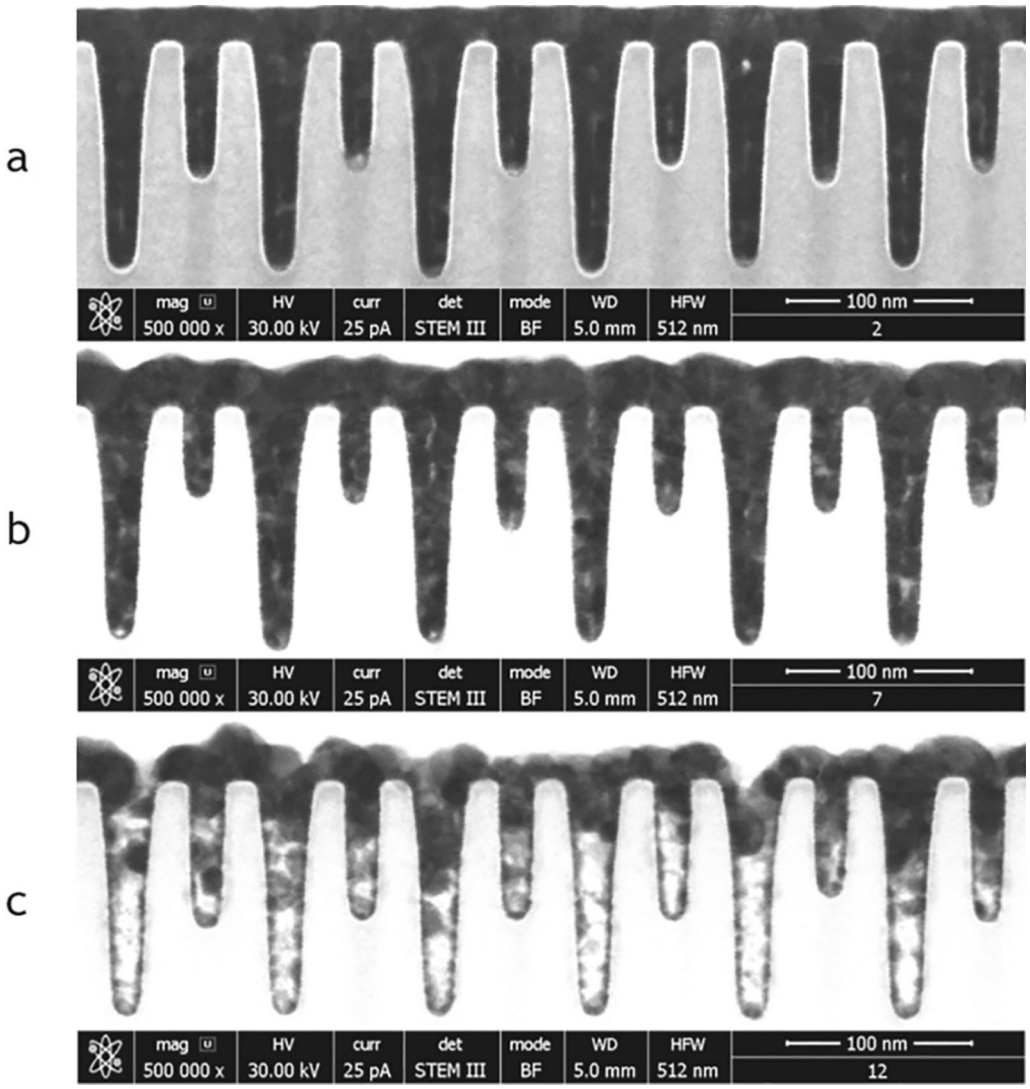


Fig. 7. STEM cross-sectional images of cobalt filling with different H_3BO_3 content: (a) 30 g/L, (b) 15 g/L, and (c) 5 g/L. Reprinted with permission from Ref. 39, Copyright 2020, IOP Publishing

not mean that alkynol/enol compounds cannot achieve cobalt bottom-up superfilling without voids. If these alkynol/enol compounds can control the rate of cobalt electrodeposition at the bottom of the via to be higher than that at the mouth, then these alkynol/enol compounds are suitable additives and meet the requirements of bottom-up superfilling.

S-Shaped Negative Differential Resistance Mechanism

Oxime compounds are condensation compounds of hydroxylamine and aldehydes or ketones, which can inhibit cobalt electrodeposition and have the characteristics of potential and diffusion dependence. Huang et al.[33,44] reported the effects of three oxime compounds, dimethylglyoxime (DMG), cyclohexane monoxime (CHO), and cyclohexane dioxime (CHD), on cobalt bottom-up superfilling. The structures of DMG, CHO, and CHD are presented in Fig. 9a. At low overpotentials, all three additives have an inhibiting effect on cobalt electrodeposition. At high overpotentials, the three additives lose their inhibiting ability, and, among them, DMG is more likely to be inactivated at a more positive potential. When a more negative potential (large current) is applied, oxime compounds lose their activity. At the same time, since oxime compounds also have mass transfer and diffusion dependence, the oxime compounds at the via mouth are continuously replenished but cannot be replenished in time at the bottom. In other words, the cobalt electrodeposition rate is faster at the bottom, and, therefore, cobalt bottom-up superfilling can be achieved. Based on this characteristic of potential and diffusion dependence, an S-shaped negative differential resistance (S-NDR)

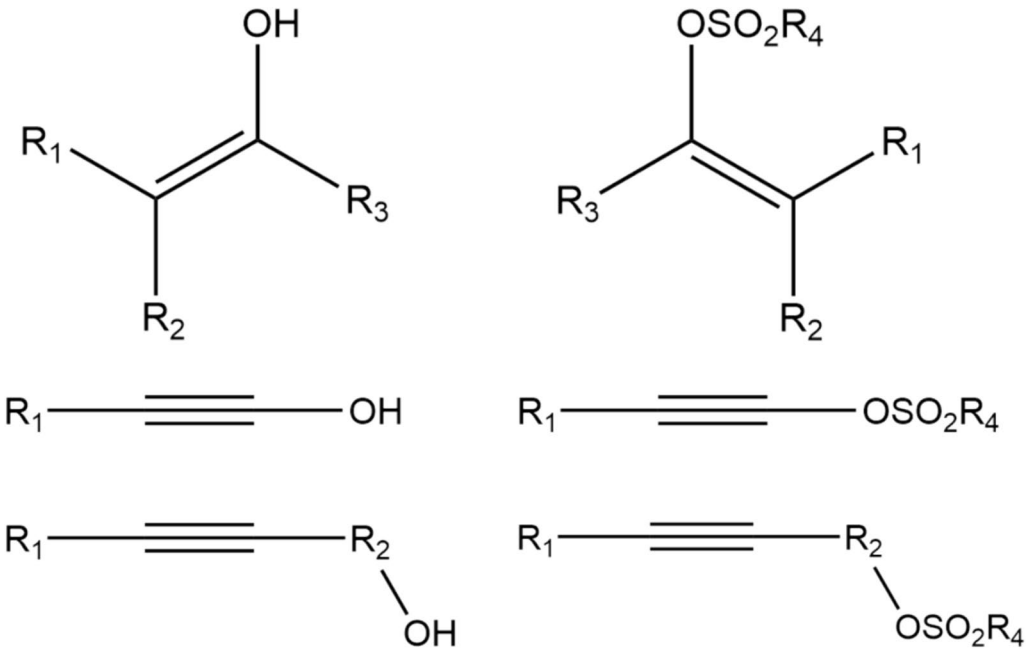


Fig. 8. Molecular structures of alkynol/enol compounds and their sulfonated compounds (R_1, R_2, R_3 and R_4 can be hydrogen or a hydrocarbon group).

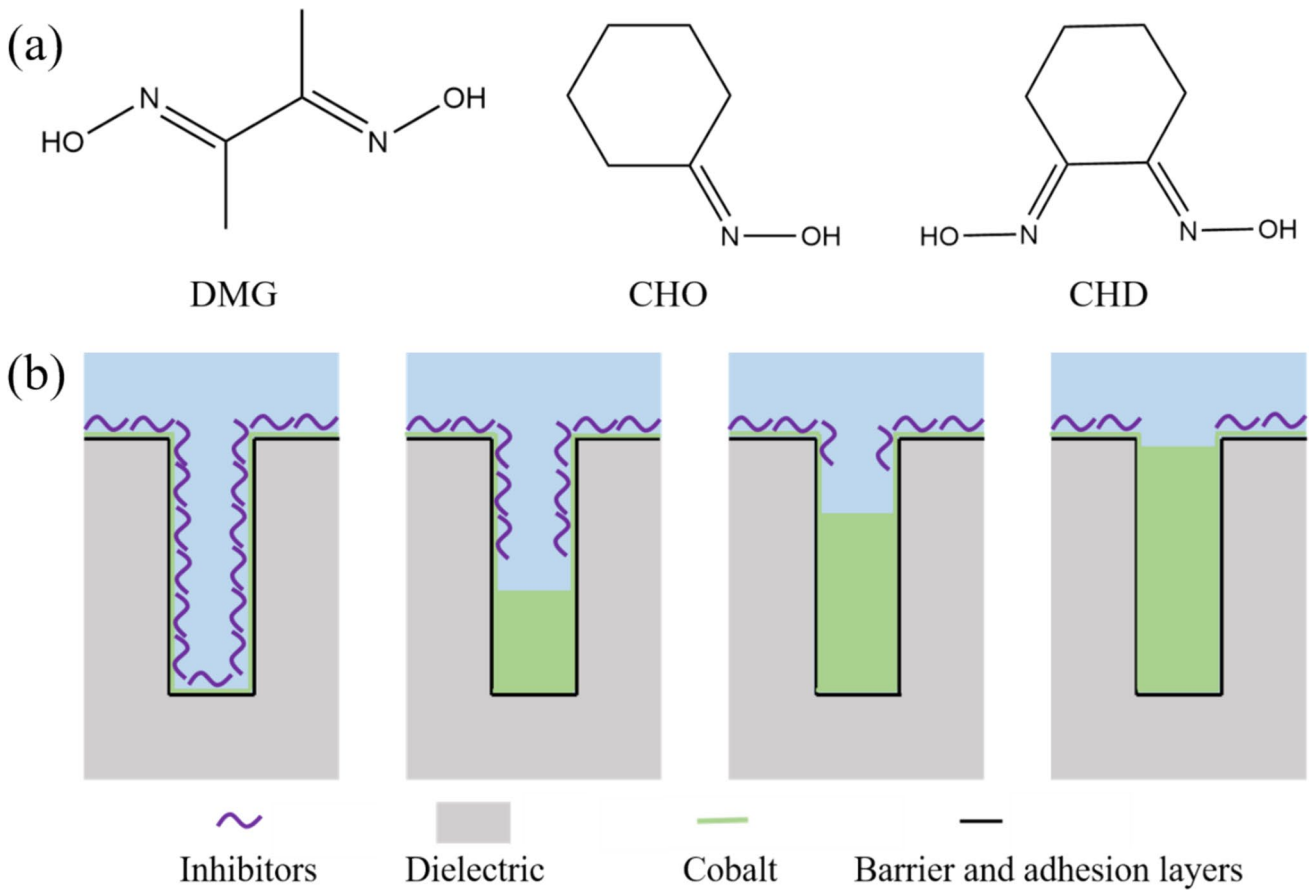


Fig. 9. (a) Structures of DMG, CHO, and CHD, (b) schematic of the S-NDR mechanism for cobalt bottom-up superfilling.

mechanism suitable for cobalt bottom-up superfilling is proposed, as presented in Fig. 9b.

Diffusion-Dependent and Induction–Diffusion–Adsorption Mechanism

Dye additives are usually nitrogen-containing compounds, which are often regarded as levelers and applied to copper electroplating interconnects.[45–54] Dye additives have the characteristics of inhibiting metal electrodeposition and diffusion dependence, so they can also be used for cobalt electroplating interconnects. Nitrotetrazolyl blue (NBT) and malachite green (MG) are the typical dye-type additives for cobalt electroplating, and their corresponding structures are shown in Fig. 10a. Li et al.[31] showed that NBT has a concentration gradient in the trench structure. As cobalt electrodeposition proceeds, the NBT content at the bottom of the trench is lower than that at the mouth, and therefore cobalt can be filled from bottom to top. Figure 10b shows the proposed diffusion-dependent mechanism for cobalt bottom-up superfilling based on the NBT action mechanism. It is worth noting that, unlike copper electroplating, the bottom-up superfilling mechanism of cobalt electroplating is also related to the atomic hydrogen concentration on the electrode surface in addition to the type of additives. Some dye additives can promote the occurrence of the hydrogen

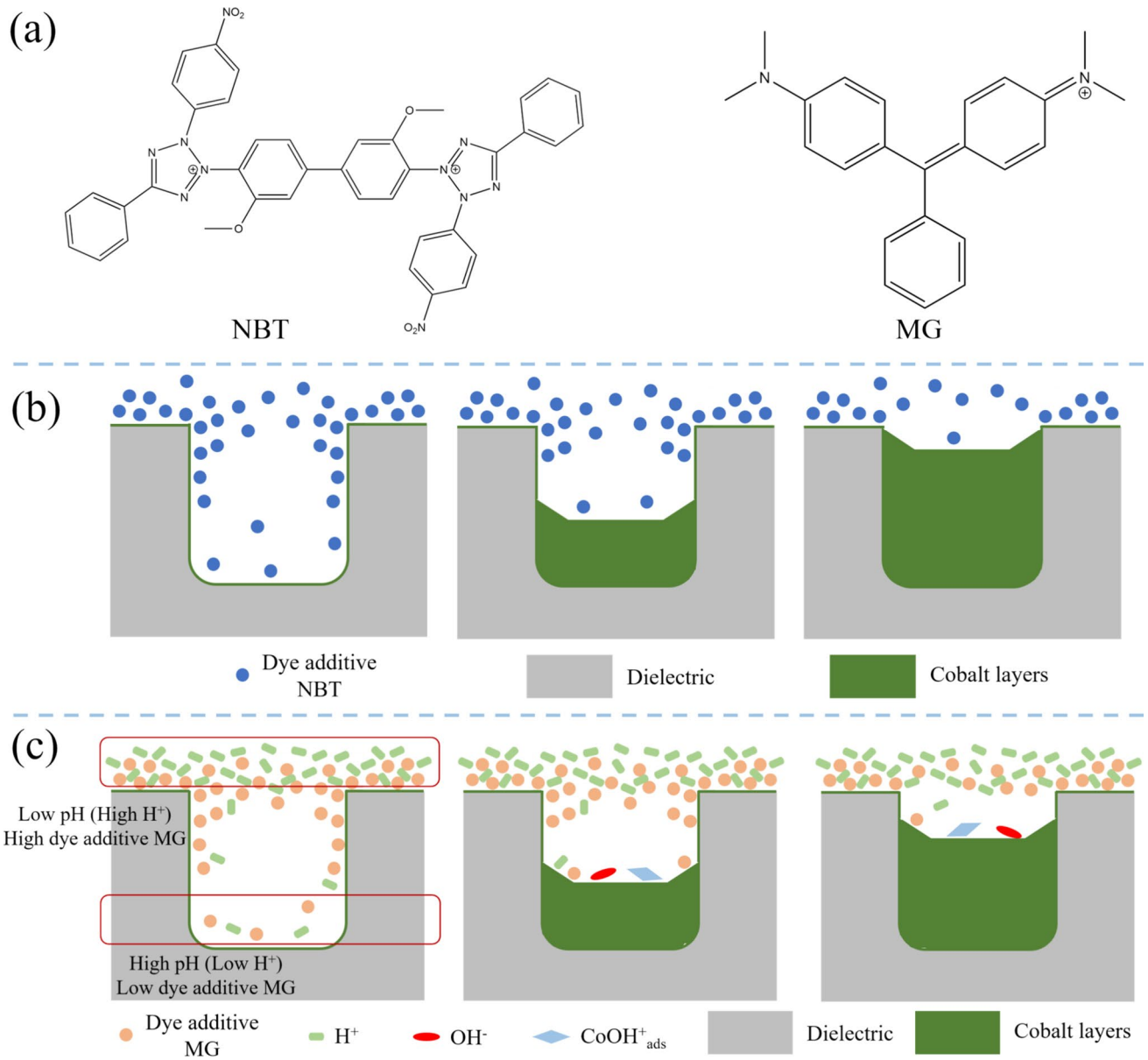


Fig. 10. (a) Structures of NBT and MG, and schematics of (b) diffusion-dependent and (c) induction–diffusion–adsorption mechanism.

evolution reaction, which affects the bottom-up superfilling effect by changing the cobalt electrodeposition current efficiency. Ma et al.[55] believed that MG can promote the generation of atomic hydrogen. There was a concentration gradient of MG in the trench, and the MG concentration was high at the trench mouth. At the same time, MG promoted the occurrence of the hydrogen evolution reaction at the trench mouth, resulting in a decrease in the current efficiency of cobalt electrodeposition there. In contrast, the current efficiency of cobalt electrodeposition at the bottom of the trench was high. Therefore, cobalt was densely superfilled from bottom to top. Based on this characteristic, an induction–diffusion–adsorption mechanism suitable for cobalt bottom-up superfilling was proposed, as shown in Fig. 10c.

Differential Current Efficiency Fill Mechanism

The above-mentioned cobalt bottom-up superfilling mechanism is closely related to the kinds of additives, but there are also some mechanisms that do not limit the kinds of additives, such as the differential current efficiency fill (DCEF) mechanism. Rigsby et al.[56–58] considered that the local pH changes directly affected the concentration of hydroxide ions involved in the formation of intermediates, $Co(OH)_{ads}$, thereby increasing the reaction paths, changing the thermodynamics of cobalt electrodeposition, and adjusting the rate of cobalt electrodeposition. They confirmed that there was a concentration gradient of hydrogen ions in the via (high concentration at the mouth of via), and the current efficiency of cobalt electrodeposition at the bottom of the via was four times higher than that at the mouth by using cyclic voltammetry with electrochemical quartz microbalance technology. The formation of a pH gradient in the longitudinal direction of the via structure led to differences in the cobalt electrodeposition rate and efficiency, which enabled bottom-up superfilling. Based on the experimental results, the DCEF mechanism was proposed and is shown in Fig. 11.

Hydrogen Reduction-Induced Deactivation

Another mechanism that is not affected by the types of additives is the hydrogen reduction-induced deactivation (HID) mechanism. Wu et al.[59,60] verified that the change of pH value affected the inhibiting ability of additives through cyclic voltammetry. During the cobalt electrodeposition process, a large amount of atomic hydrogen accumulates at the bottom of the via due to the large surface-to-volume ratio, and the inhibitor at the bottom of via is continuously protonated and loses the ability to inhibit the electro-reduction of cobalt ions. A schematic of the HID mechanism is presented in Fig. 12. According to the principles described in this mechanism, any additive that undergoes a transition between an inactive state and active state under the influence of hydrogen ions can control and achieve the cobalt bottom-up superfilling.

Comparison and Applicability of Mechanisms

For the S-NDR mechanism, it is often believed that whether or not the deposition is suppressed in metal superfilling is related to the adsorption and deactivation of the suppressant in the bath. If there is a suitable convection diffusion gradient in the depth direction of the trench structure, cobalt filling from bottom to top can be achieved. This mechanism is limited by the type of additives, and oxime compounds often have this characteristic. For the diffusion-dependent and induction–diffusion–adsorption mechanisms, these are closely related to the hydrogen evolution reaction and diffusion. However, dye molecules are often unstable and are easily decomposed during the electroplating process. Their

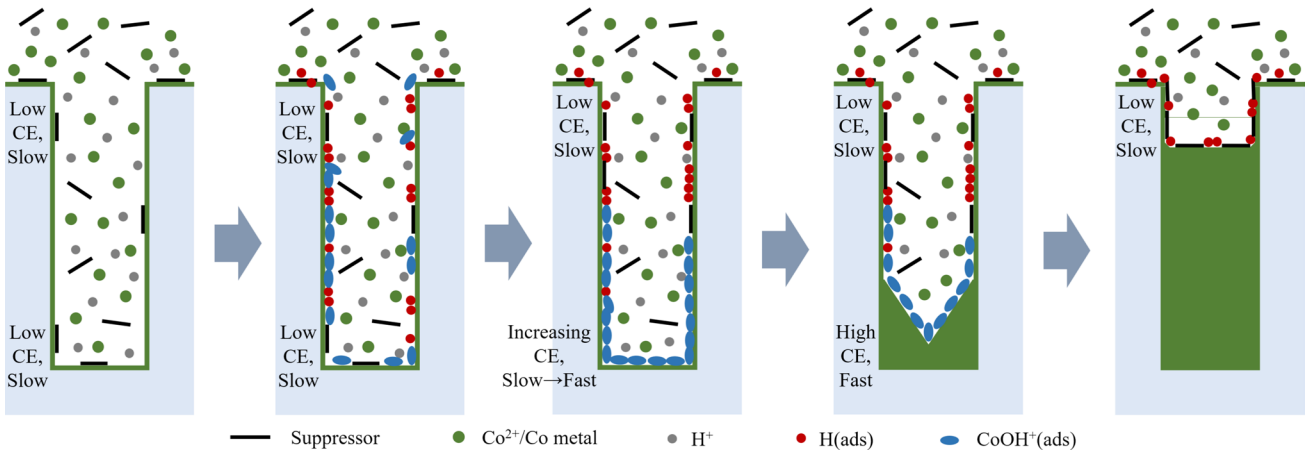


Fig. 11. Schematic of DCEF mechanism. Adapted from Ref. 56, Copyright 2019, IOP Publishing, under the terms of the Creative Commons Attribution 4.0 License (CC BY, http://creativecommons.org/licenses/by/4.0/)

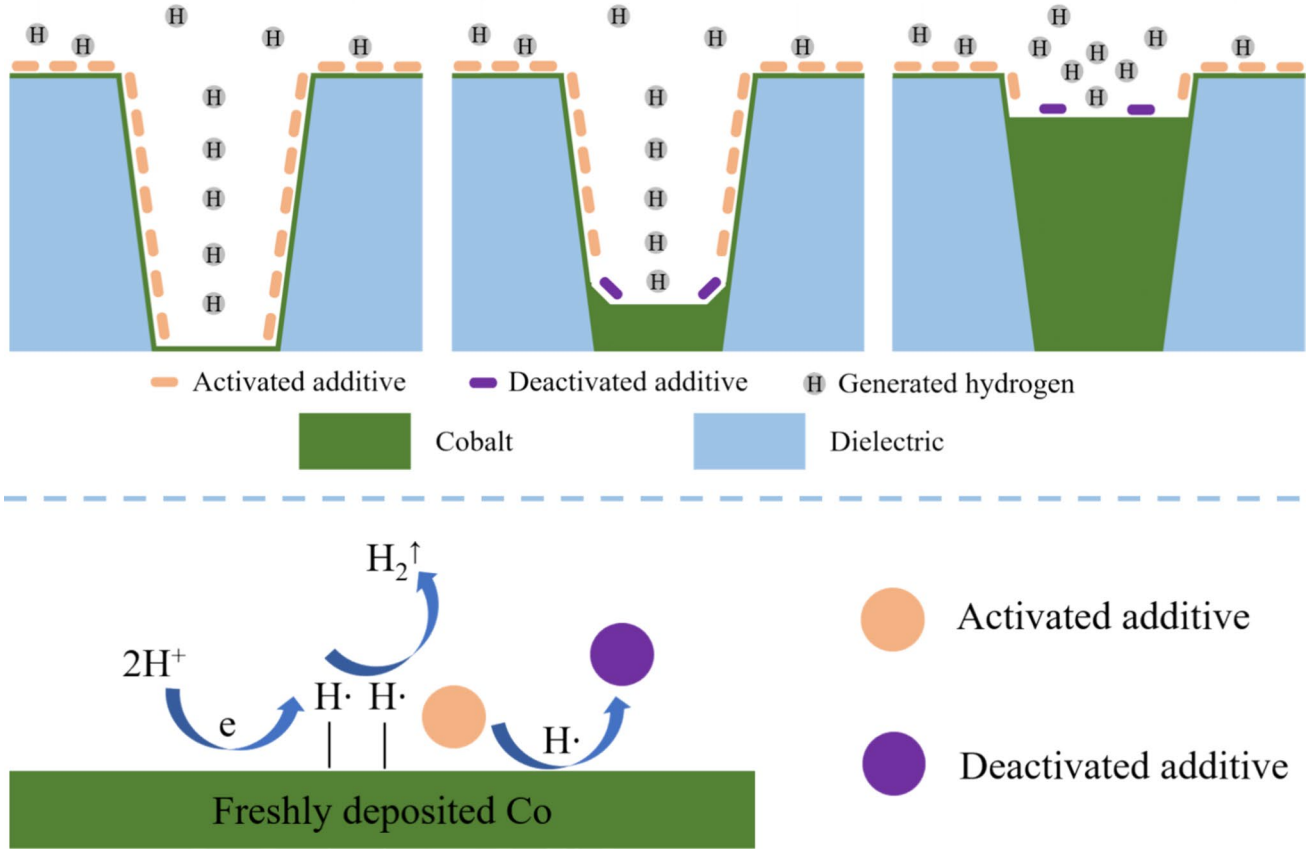


Fig. 12. Schematic of HID mechanism. Adapted from Ref. 59, Copyright 2019, IOP Publishing under the terms of the Creative Commons Attribution 4.0 License (CC BY, http://creativecommons.org/licenses/by/4.0/)

effect cannot last long, and the decomposition products may have a negative impact on the cobalt electrodeposition process. For the DCEF mechanism, this mechanism does not care about the type of additives. As long as a pH gradient can be formed in the trench structure, it can affect the cobalt electrodeposition rate and current efficiency at the bottom and the mouth, making the current density of cobalt electrodeposition at the bottom much higher than that at the mouth. However, to form a suitable pH gradient, the electroplating current density needs to be controlled below the limiting diffusion current density of proton reduction in order to prevent the proton consumption rate from being much greater than the mass transfer rate and failing to achieve superfilling. For the HID mechanism, this mechanism also does not care about the type of additive. The specific surface area at the bottom of the trench is large, and more atomic hydrogen gathers here. As long as the additive can be reduced and inactivated by atomic hydrogen, superfilling can be achieved. Since the local current density at the bottom exceeds the current density at the mouth, affected by the current efficiency, when the electrodeposition approaches the mouth, the local current density is still much greater than the local current density near the mouth. The current density effect will still provide more hydrogen in the feature area, and superfilling continues to occur, resulting in mushroom-shaped bulges. The above-mentioned cobalt superfilling mechanisms have different emphases, and sometimes two bottom-up filling mechanisms may exist simultaneously in a bath.

Surface and Interface Behavior of Cobalt

The surface and interface formed by the electrode and the bath are important reaction sites in the cobalt electroplating process. Different from the traditional conventional electroplating, cobalt electroplating interconnects are carried out in the micro-/nanovia or trench with a certain aspect ratio, in which surface and interface structure, potential distribution, metal nucleation and growth mechanism, mass transfer process, additive molecular action mechanism, etc. are more complex. Therefore, it is significantly important to clarify the structure of the surface and interface of electroplating and the mechanism of cobalt electrodeposition to accelerate the fundamental research and improve development efficiency.

Electro-reduction

Cobalt ion is a divalent metal ion. In the inherent impression, divalent cobalt is directly electro-reduced to metal cobalt, and the chemical formula is: $Co^{2+} + 2e = Co(s)$.[61,62] However, divalent cobalt ions, like iron group elements, need to go through an intermediate state for electro-reduction.[63–68] The electro-reduction reaction of cobalt ions proceeds in a continuous one-electron transfer step through the adsorbed cobalt hydroxide intermediate, and the corresponding chemical formulae are shown below.[56] It is worth noting that Rigsby et al.[56] confirmed that the rate of cobalt electro-reduction varies depending on the concentration of hydroxide ions.

$$Co^{2+} + OH^- = \left(CoOH^+\right)_{ads} \quad (1)$$

$$\left(CoOH^+\right)_{ads} + e = (CoOH)_{ads} \quad (2)$$

$$(CoOH)_{ads} + e = Co(s) + OH^- \quad (3)$$

Some researchers have different views on the electro-reduction of cobalt ion. Santos et al.[69] deemed that the cobalt ions combine with the hydroxide to form cobalt hydroxide, which is a competitive reaction with the cobalt ions being directly electro-reduced to metal cobalt. The formed cobalt hydroxide is adsorbed on the electrode surface, and obtains two electrons at a relatively negative potential to be electro-reduced to metal cobalt. The corresponding chemical formulae are:

$$Co^{2+} + 2OH^- = \left(Co(OH)_2\right)_{ads} \quad (4)$$

$$\left(Co(OH)_2\right)_{ads} + 2e = Co(s) + OH^- \quad (5)$$

Nucleation and Growth

The nucleation and growth process of cobalt interconnect wires has a huge impact on their stability and resistivity. Pagnanelli et al.[70] found that increasing the overpotential of cobalt electrodeposition will increase the number of active sites on the surface, resulting in an increase in the density of cobalt nuclei and a significant decrease in grain size. Mehmood et al.[71] confirmed that the overpotential of cobalt electrodeposition is closely related to its growth structure. High overpotential is conducive to the formation of face-centered cubic cobalt clusters with a small core radius. In addition, pH has an effect on the nucleation of micron-scale cobalt, but has no effect on the nucleation of nanoscale cobalt.[72] Mukhtar et al.[73] showed that the concentration of cobalt ion has a significant effect on its nucleation. When the concentration is low, the cobalt nucleus is larger and the structure is hexagonal close-packed. As the concentration is high, the cobalt nucleus is smaller and the structure is face-centered cubic. A schematic is shown in Fig. 13.

The presence of additives can also affect the nucleation behavior of cobalt. Hu et al.[34] reported the nucleation and growth process of cobalt under the action of different additives (DMG, CHD). As shown in Fig. 14, there was a two-step nucleation process on the electrode surface, including electro-reduction of cobalt coordination ions (formed by additives and cobalt ions) and direct electro-reduction of cobalt ions. Furthermore, Hu et al.[74] also discovered that the threshold potential for nucleation decreased upon the increase of furil dioxime concentration, and the deposition was found to follow the progressive nucleation behavior for almost all cases.

Electrochemical Oscillation

The phenomenon that an autocatalytic reaction causes a positive feedback loop is regard as electrochemical oscillation.[75–78] The self-adsorption/desorption, decomposition and competitive adsorption of additives on the electrode surface may change the kinetic mechanism of metal ion electro-reduction, resulting in electrochemical oscillation, and periodic changes in current or potential within a certain range.[78–81] Numerous studies on electrochemical oscillation for copper interconnects have been reported,[82–85] but, to the best of our knowledge, only one report on the electrochemical oscillation of cobalt interconnects has been published. Hu et al. [86] reported that 3-mercapto-1-propanesulfonate (MPS) can lead to an electrochemical oscillation. The main positive feedback loops resulting in oscillation are autocatalytic accumulation and dissolution of cobalt hydroxide, which are triggered by the potential-dependent adsorption and desorption of the Co-MPS complex. Figure 15 shows the potential oscillation phenomen0n and mechanism of cobalt.

Coordination

Additive molecules can coordinate metal ions on the electrode surface, thereby changing the electrode process kinetics of metal deposition. For example, in copper interconnects, MPS can coordinate with copper and chloride ions to form a complex, thereby accelerating copper electrodeposition.[87,88] Polyethylene glycol can also coordinate with copper and chloride ions to form a complex to inhibit copper electrodeposition, [89,90] while cobalt ions can also coordinate with additives during electro-reduction. Li et al.[31] showed that a coordination interaction is formed between the opened nitrogenous heterocyclic ring in NBT and cobalt ions, as the cobalt ions move toward the electrode surface.

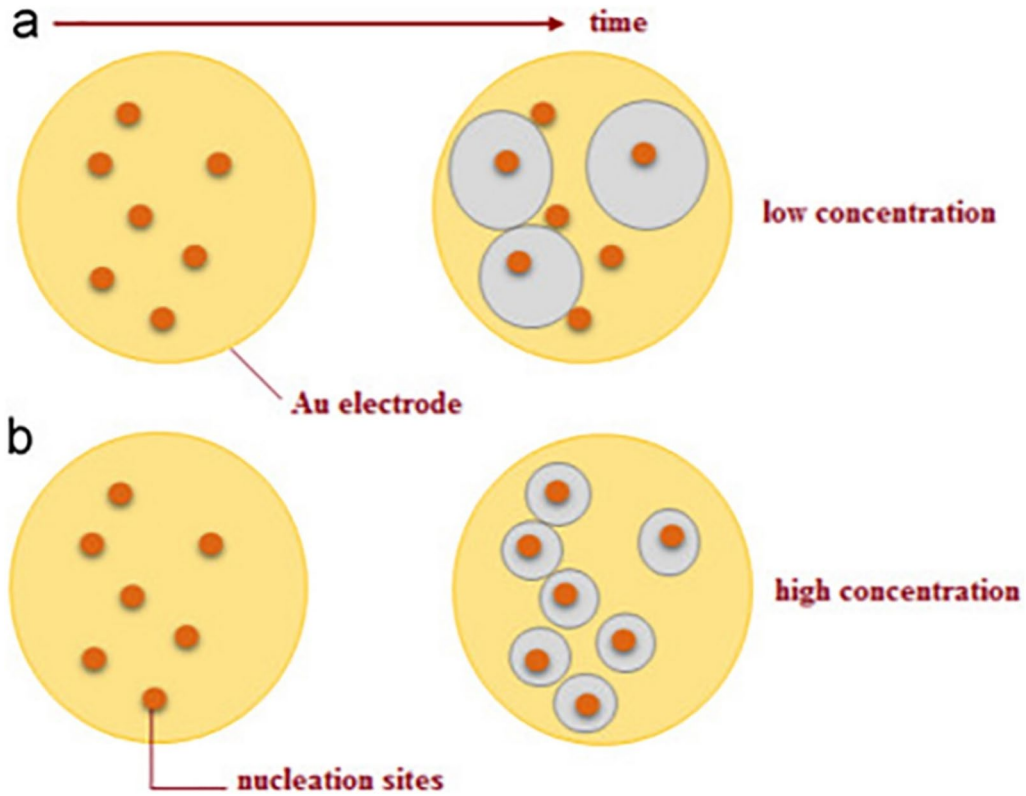


Fig. 13. Schematic of the effects of concentration of cobalt on nucleation. Reprinted with permission from Ref. 73, Copyright 2016, Elsevier.

This complex can inhibit cobalt electrodeposition through synergistic adsorption and coordination effects. Ma et al.[91] showed that added malachite green can coordinate with cobalt ions to increase the electrochemical polarization by the adsorption–coordination–inhibition mechanism, as shown in Fig. 16.

Microstructure of Cobalt Electroplating

Electroplating parameters have a profound and complex impact on the microstructure of cobalt, which is directly related to the performance of the cobalt coating and determines its value in actual applications. In electroplating production, it is necessary to comprehensively consider the parameters, such as current density, temperature, concentration, additives, etc., and obtain the ideal cobalt microstructure through precise process control, so as to meet the unique requirements of various application fields for the performance of cobalt coatings. Pan et al.[43] showed that, in the presence of saccharin, tetrabutylammonium bromide slightly affects the morphology, microstructure, surface roughness, grain size, and microhardness of cobalt deposits, while 2-butyne-1,4-diol can transform the morphology of the deposits from a fine-grained structure to a lenticular structure, changing the microstructure from a (0002) texture to a (1010) texture, and significantly reducing the surface roughness and microhardness. By adding a combination of saccharin, tetrabutylammonium bromide, and 2-butene-1,4-diol, a harder and smoother cobalt deposit can be obtained. Mahdavi et al.[92] reported that increasing the concentration of cobalt sulfate and sodium dodecyl sulfate in the bath has a negligible effect on the microhardness of the coating, but has an effect on the electrodeposition current efficiency. Adding saccharin to the electrodeposition bath can reduce the grain size of the hexagonal close-packed cobalt film and increase its microhardness, but has no significant effect on the current efficiency. Wu et al.[93,94] reported the effect of phenylmethyl sulfoxide (PMSO) on cobalt coatings, and found that an appropriate PMSO concentration (about 20 ppm) is important for achieving a low surface roughness of about 3.9 nm and a unit hcp (002) orientation, as shown in Fig. 17. Under the action of PMSO, the nucleation of cobalt is transformed from instantaneous nucleation to progressive nucleation, resulting in a uniform deposition behavior of cobalt. The presence of PMSO can also enhance the wettability of the plating solution. In addition, it was found that the electron-donating ability of the sulfite group in benzenesulfinic acid (BNS) is higher than that of the sulfonic acid group, resulting in stronger adsorption of

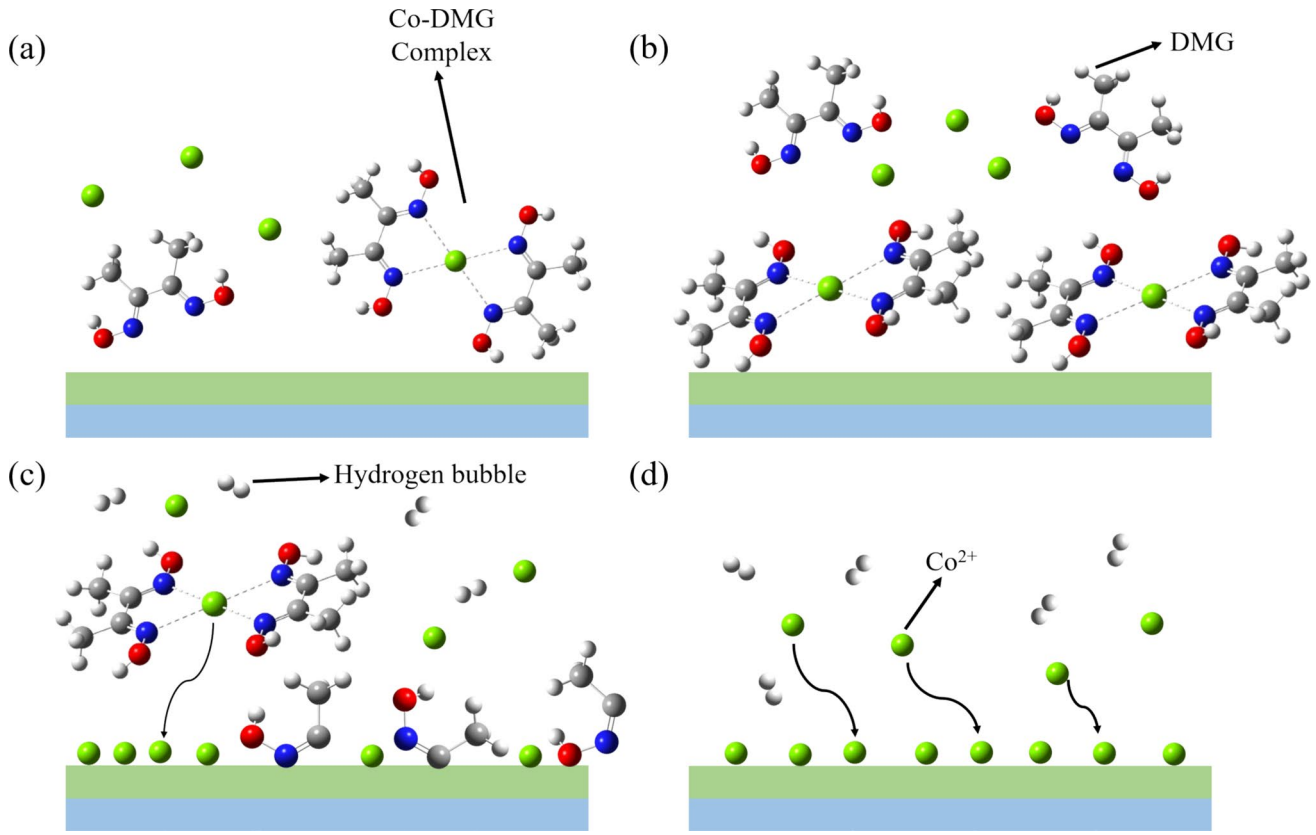


Fig. 14. Nucleation mechanism of cobalt under the action of additives. Adapted from Ref. 34, Copyright 2018, IOP Publishing under the terms of the Creative Commons Attribution 4.0 License (CC BY, http://creativecommons.org/licenses/by/4.0/).

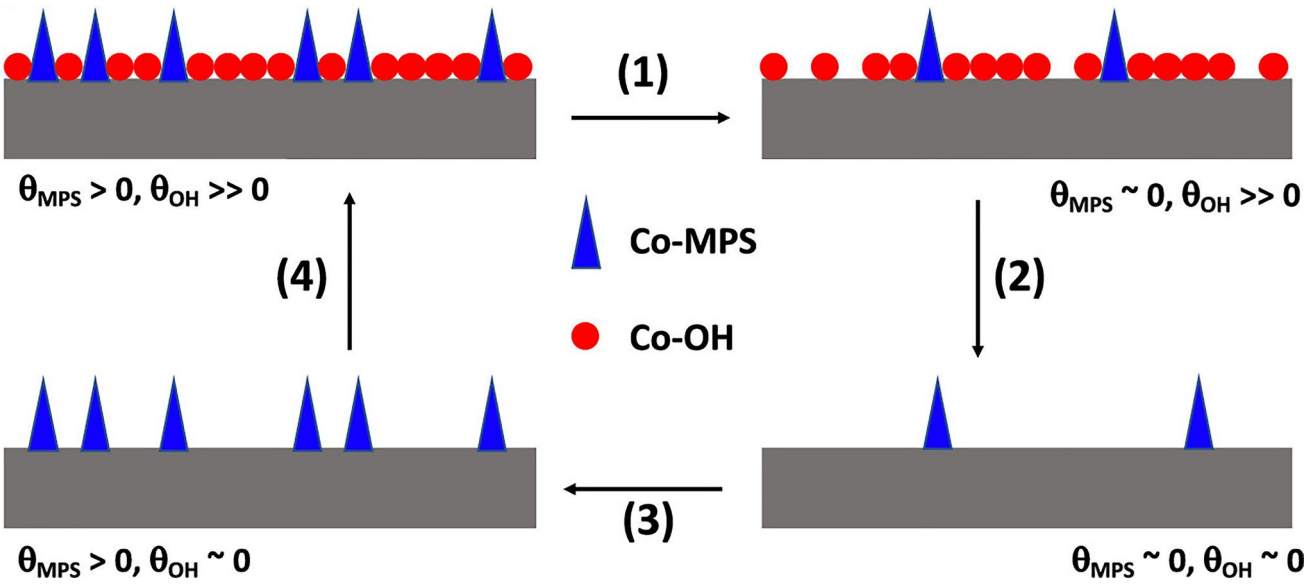


Fig. 15. Potential oscillation phenomenon and mechanism of cobalt. Reprinted with permission from Ref. 86, Copyright 2020, American Chemical Society.

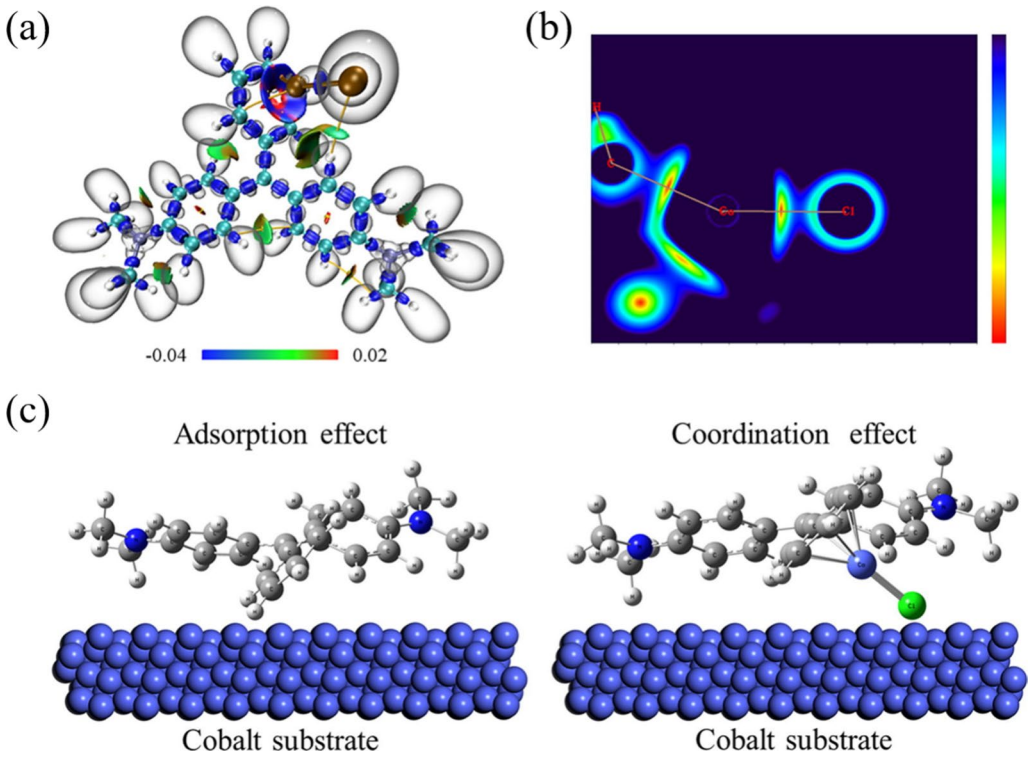


Fig. 16. Quantum chemical calculation and adsorption–coordination–inhibition mechanism. Reprinted with permission from Ref. 91, Copyright 2022, Elsevier.

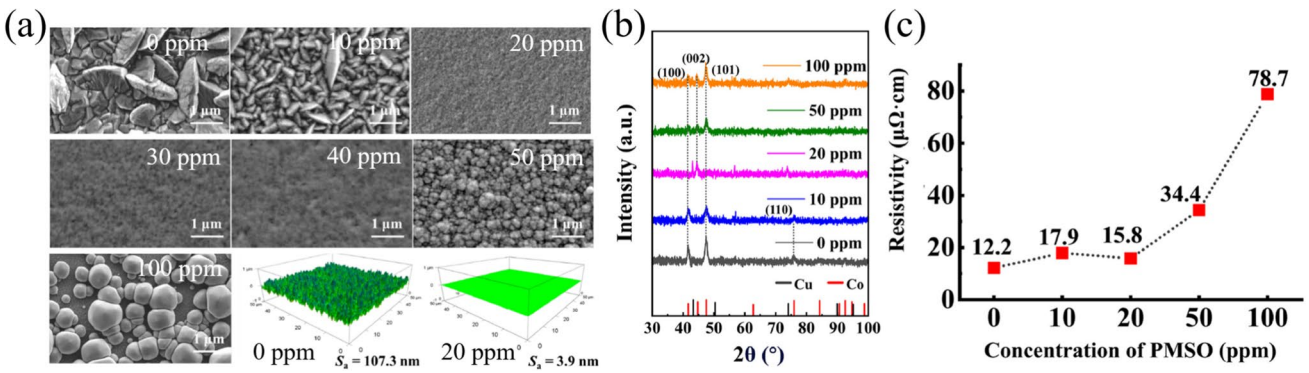


Fig. 17. (a) Scanning electron microscope images of cobalt films electrodeposited from plating baths with different concentrations of PMSO, (b) X-ray diffraction results of electrodeposited Co films prepared from solutions containing different concentrations of PMSO, and (c) The resistivity of the electrodeposited cobalt films. Reprinted with permission from Ref. 94, Copyright 2025, Elsevier.

BNS on the cobalt surface and enhanced cathodic polarization. BNS also transforms the nucleation of cobalt from instantaneous nucleation to progressive nucleation, resulting in a more uniform and dense cobalt film morphology. The selectivity and strong adsorption of BNS on the cobalt plane are the reasons for the preferential orientation growth of the (002) crystal plane. After annealing, the cobalt film prepared by BNS had obvious grain growth, obvious impurity diffusion, and further improved electrical properties. Su et al.[95] showed that the hardness of nanocrystalline Co coatings increases with the decrease of grain size. Small average grain sizes and smooth surfaces contribute to the improvement of corrosion resistance of the coatings. A coating obtained by reverse pulse electrodeposition had the finest grains, the highest hardness, and the highest corrosion resistance, followed by bipolar pulse electrodeposition. Hardness, surface roughness, morphology, and phase structure have significant effects on the tribological properties and wear mechanism of nanocrystalline Co coatings, with those prepared by dual pulse electrodeposition showing the highest

wear resistance and the lowest friction due to their higher hardness and smoothness. Nanocrystalline Co coatings prepared by reverse pulse electrodeposition showed the highest friction and the lowest wear resistance due to their higher surface roughness, the absence of an hcp(110) peak, and the weakening of the hcp(100) peak.

Potential Future Direction

At present, the study of cobalt electroplating often obtains average information of various thermodynamics and kinetics of the electrode/solution interface and electrode surface processes at the macrolevel by using classical electrochemical research methods and theoretical calculations. However, classical electrochemical research methods find it difficult to identify the characteristics of electrode reaction intermediates and products, and cannot provide mechanism information at the microscopic and molecular levels. Some advanced in situ characterization methods are used in copper electroplating.[96–99] Therefore, advanced electrochemical in situ research methods should be used in the future. In addition, annealing treatment of cobalt coatings is also a future research direction.

1. Electrochemical in situ infrared spectroscopy can reveal the adsorption, orientation, bonding, dissociation, and other behaviors of species at the electrode/electrolyte interface, as well as the real-time changes of various species during the electrochemical reaction.[90,100–102] Since the Raman signal of water molecules is very weak, electrochemical in situ surface-enhanced Raman spectroscopy has high sensitivity, high signal-to-noise ratio and spatial resolution, which is conducive to studying the molecular mechanism of additives.[103–105] At the same time, Raman spectroscopy has been combined with digital imaging technology to develop a Raman mapping technology, which can obtain the spatial distribution information of specific spectral characteristics of Raman-active molecules.[106,107]
2. The nucleation and growth process of cobalt electrocrystallization determines the quality and performance of the cobalt coating.[108] Since the cobalt nucleation and growth process is very rapid, the crystal nuclei grown at the solid/liquid interface are submicron/nanoscale, which brings certain difficulties to in situ research. In situ transmission electron microscopy or in situ scanning electron microscopy) has high spatial resolution and are the most intuitive and efficient research methods for studying nucleation and growth.[109–111] In situ observation, recording the nucleation and growth behavior of the initial stage of metal deposition, and studying the influence of additives on the cobalt nucleation process and growth, can provide theoretical guidance and technical support for electroplating, especially chip electroplating.
3. X-ray diffraction (XRD) is an effective means of analyzing the structure of the coating, and can obtain the properties of the coating, such as atomic arrangement, preferred crystal orientation, grain size, and microstress.[112,113] These properties determine the physical properties of the interconnects, such as conductivity, thermal conductivity, and ductility, and thus determine the quality of the interconnects. Electrochemical in situ XRD technology can monitor the dynamic evolution of coating composition, crystal face, and structure in real time, and study the mechanism of the influence of plating solution composition on the properties of the coating. It is crucial to reveal the structure–activity relationship between the structure and physical properties of the coating and to establish an interconnect quality evaluation mechanism.
4. The introduction of impurity atoms affects the grain structure and preferred orientation of cobalt, and its redistribution and segregation in the coating will also affect the change of resistance.[114,115] Like copper electroplating, the electroplated cobalt also needs to be annealed and recrystallized to allow the impurities in the coating to escape and thus improve its conductivity.

Conclusions

Electroplating is a critical technology in high-end electronic manufacturing, including integrated circuits. As interconnect dimensions continue to shrink, copper electroplating is increasingly unable to meet the required performance and safety standards. Cobalt, with its superior electrical properties and safety at the micro-/nanoscale, is emerging as a promising next-generation interconnect material to replace copper. Although there have been reports on cobalt electroplating, current research has primarily focused on the electrochemical behavior of cobalt ions and additives, the role of additives, and the filling effects in trenches and vias, along with proposed filling mechanisms. However, several key aspects remain insufficiently explored. These include the specific actions and synergistic/antagonistic interactions of components such as boric acid and additives in the plating bath, the mechanisms by which pH, temperature, and individual bath components influence cobalt nucleation and growth, and the surface, interface, and filling processes during cobalt electrodeposition in micro-/nanostructures. Future research directions should prioritize the development and application of advanced in situ techniques, including morphological, spectroscopic, and energy spectroscopic, to address these challenges. Clarifying these mechanisms will

be essential for advancing cobalt electroplating as a viable interconnect technology in next-generation chip fabrication.

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Data Availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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